
Prediction Defect Equalization for Fast Diffusion Sampling

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Denoising diffusion models generate high-quality images by simulating a reverse
2 trajectory from noise to data, but this iterative procedure becomes costly when
3 many denoiser evaluations are required. A central challenge in fast sampling is
4 therefore to choose a discretization schedule that places a small computational
5 budget at the most important parts of the reverse trajectory. Existing schedules are
6 often hand-crafted or optimized using indirect criteria such as distributional bounds,
7 transport costs, or information increments. These criteria can produce useful grids,
8 but they do not directly reveal where a particular numerical solver makes its most
9 consequential local errors along the denoising path. This paper presents *geometric*
10 *prediction defect equalization*, a training-free algorithm for automatically selecting
11 sampling schedules for deterministic diffusion ODE samplers. Our method uses
12 accurate teacher trajectories from a pretrained model to estimate where a target
13 solver locally deviates from the intended trajectory, and then allocates time steps
14 to equalize these geometric prediction defects. The resulting schedule is solver-
15 aware, coordinate-covariant, and requires no model retraining, sampler distillation,
16 or tuning of quality-based objectives. We show that, under a local power-law
17 defect model, our schedule is asymptotically minimax-optimal for controlling the
18 maximum local distributional prediction defect. In validated PNDM/CIFAR-10
19 Euler 50k-sample runs, ours improves FID over base, native-coordinate linear, and
20 an authorized project-owned offline-proxy schedule at NFE 10 and 20 across three
21 seeds. Comparisons to fixed Karras/EDM-style grids and matched SD1.5/SDXL
22 AYS schedules remain mixed, highlighting the current limits of defect-based
23 schedule calibration.

1 Introduction

25 Denoising diffusion models generate images by simulating a reverse trajectory from noise to data.
26 This simulation is accurate when many denoiser evaluations are available, but those evaluations
27 dominate sampling cost. Fast sampling therefore asks a concrete allocation question: given only a
28 small number of function evaluations, where along the reverse trajectory should they be placed? The
29 answer is not determined by the solver alone. Even a strong numerical update can fail if its few steps
30 are spent in the wrong regions of the denoising path.

31 Discretization schedules are the mechanism for making this allocation. Linear, polynomial, log-SNR,
32 and EDM-style schedules are simple, stable, and widely used [Karras et al., 2022], but they are
33 usually hand-crafted and reused across models, solvers, and budgets. Recent work shows that this
34 choice can be optimized. AYS searches for schedules with a solver-specific Girsanov KL upper bound
35 [Sabour et al., 2024]. Score-Optimal Diffusion Schedules derive a local path cost from the work
36 performed by a score-based corrector and choose an inverse path-length schedule [Williams et al.,

2024]. Other optimized-timestep methods minimize discrepancies between ODE teacher solutions and numerical solver trajectories [Xue et al., 2024]. These approaches make clear that schedule selection is a central component of fast diffusion sampling.

However, the criteria used to choose a schedule often remain indirect from the perspective of the sampler that will actually be run. A distributional bound, transport-style cost, or path information increment can justify an effective grid. These criteria leave open the diagnostic question we care about: where does this numerical solver make its largest local prediction errors along the denoising trajectory? Without that information, schedule selection still looks like global search instead of targeted correction of the solver’s failure modes.

We present *geometric prediction defect equalization*, a training-free algorithm for automatically selecting schedules for deterministic diffusion ODE samplers. Our method uses accurate teacher trajectories from a pretrained model to estimate where a target student solver locally deviates from the intended trajectory. It then allocates time steps so that these local geometric prediction defects are equalized across the schedule.

The details of our method are designed to make this diagnostic meaningful. Each defect probe starts from the correct teacher state, so it measures one-step solver error rather than error accumulated from previous approximate steps. Teacher trajectories are treated as Monte Carlo samples from the teacher marginal at each noise level, so the estimated defect profile is distributional rather than path-specific. A state-space metric and optional tangent downweighting turn endpoint residuals into geometric prediction defects. Finally, a local power-law model determines how to convert these defects into a monitor density whose inverse CDF yields the schedule.

Our contributions are as follows.

1. **A solver-facing schedule objective.** We formulate low-NFE schedule selection as distributional local prediction-defect equalization for a fixed pretrained model and deterministic sampler.
2. **A training-free calibration algorithm.** We estimate local oracle-start defects from teacher trajectories, aggregate them over the teacher marginal with mean, CVaR, or mixed risk, and construct the schedule by inverse-CDF equalization of the induced monitor.
3. **Geometry and theory.** We derive the monitor $\omega_{\mathcal{R}}(u) = (\bar{a}_{\mathcal{R}}(u) + \epsilon_a)^{1/q}$ from a local power-law defect model, prove coordinate covariance of the monitor one-form, and prove asymptotic minimax optimality of the inverse-CDF schedule for local distributional risk.
4. **Scoped empirical evidence.** We report a real PNDM/CIFAR-10 smoke gate, three-seed 50k CIFAR-10 FID results including base, linear, Karras-style fixed, and authorized project-owned offline-proxy schedules, matched SD1.5 and SDXL CFG sweeps retained as text-to-image boundary evidence, and model-evaluation-equivalent calibration cost accounting.

The empirical claims are deliberately bounded. On PNDM/CIFAR-10 with the Euler solver, our schedule improves FID over base, native-coordinate linear, and the project-owned offline-proxy schedule at NFE 10 and 20. The comparison to a fixed Karras/EDM-style grid is mixed: ours wins at NFE 10 and trails slightly at NFE 20. The matched text-to-image sweeps show condition-dependent behavior: SD1.5 is negative under these metrics, while SDXL improves ImageReward against base at CFG 5.0 and 7.5 and has positive mean deltas against AYS at CFG 7.5. We therefore present our method as a way to expose and equalize solver-local trajectory defects; direct optimization of FID, CLIPScore, ImageReward, or human preference remains outside the claim.

2 Related Work

Our method sits between schedule optimization and solver design. The closest prior work asks how to choose the discretization grid; our distinction is that we estimate the local oracle-start defect of the target solver under the teacher marginal and then convert that defect into an equal-mass monitor.

Diffusion samplers and hand-designed schedules. DDPMs, DDIMs, and score-SDE samplers define reverse generative dynamics and expose a finite set of model-evaluation locations [Ho et al., 2020, Song et al., 2022, 2021]. EDM makes the design space of preconditioning, noise parameterization, and polynomial noise schedules explicit [Karras et al., 2022]. These schedules are cheap, stable,

and reusable, which makes them strong practical baselines. They are not, however, diagnostics of a particular solver’s local prediction defect.

Offline and path-geometric schedule optimization. AYS is the strongest direct schedule-optimization baseline in our setting. It uses stochastic calculus to build a Girsanov-based KL upper bound between the true generative SDE and a solver-induced linearized SDE, then searches for a schedule specific to a model, dataset, and solver [Sabour et al., 2024]. Its published Stable Diffusion schedules are therefore important baselines in the text-to-image sweep.

Score-Optimal Diffusion Schedules are especially close conceptually [Williams et al., 2024]. Williams et al. define a diffusion path cost from the work a Langevin corrector must perform after a predictor step. The resulting optimal generator is an inverse path-length parameterization: local costs are equalized along the sampling path. We keep this equalization principle but change the object being equalized. Instead of a path-intrinsic score-corrector cost under perfect-score assumptions, we measure the oracle-start one-step residual of the actual deterministic solver against a teacher flow, aggregate it over p_u , and use the q -th root dictated by the measured local defect order.

Other optimized-timestep methods also move beyond fixed grids. Xue et al. optimize time steps by minimizing the distance between a ground-truth ODE solution and the numerical solution induced by a specific solver [Xue et al., 2024]. This is close to the teacher-student motivation of our method, but their method solves an offline constrained optimization problem, whereas ours estimates a distributional local defect profile and materializes the schedule by inverse-CDF equalization.

Solver acceleration. PNDM, DPM-Solver, DPM-Solver++, UniPC, and related predictor-corrector solvers reduce sampling cost by changing the numerical update rule [Liu et al., 2022, Lu et al., 2022, 2025, Zhao et al., 2023]. STORK uses stabilized Taylor orthogonal Runge–Kutta ideas to improve mid-NFE diffusion and flow-matching sampling while addressing stiffness and structure-dependence [Tan et al., 2026]. These methods are complementary to our method: they change how a step is computed, while ours changes where a fixed solver evaluates the model. The distinction also matters for theory. Our main guarantee is stated for deterministic oracle-start ODE defects; multistep, stochastic, and scheduler-history solvers require either augmented-state analysis or empirical framing.

Learned fast samplers. Differentiable sampler search and progressive distillation reduce NFE by optimizing sampler parameters or distilling many-step samplers into fewer-step generators [Watson et al., 2022, Salimans and Ho, 2022]. Our method does not update model weights and does not optimize an image-quality objective. Negative cases therefore diagnose the schedule monitor and its calibration distribution rather than a learned generator.

3 Method

Our method turns the practical question of low-NFE sampling into a calibration problem. The input is a pretrained diffusion model, a deterministic sampler that will be used at inference, and a desired number of steps. The output is a schedule that spends more of those steps where the sampler’s own one-step prediction is locally unreliable. This section first defines the teacher and student objects, then shows how teacher trajectories are used to estimate a defect monitor and invert it into a schedule.

3.1 Problem setup

We start from a deterministic diffusion or flow ODE written in a monotone sampling coordinate u :

$$\frac{dx}{du} = f_\theta(x, u), \quad u \in [u_{\min}, u_{\max}]. \quad (1)$$

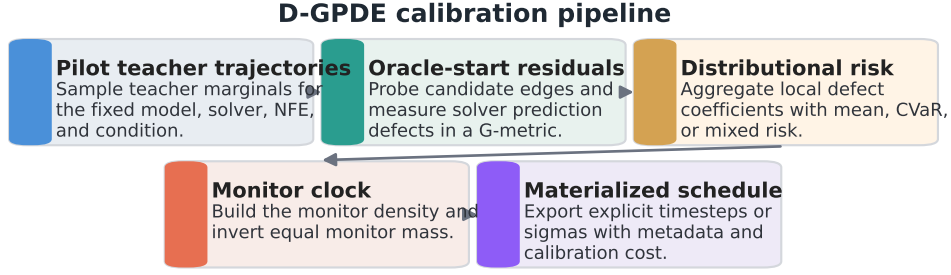
The coordinate may denote time, noise level, log-SNR, or another smooth monotone reparameterization. Let $F_{a \rightarrow b}(x)$ denote the exact teacher flow, approximated in practice by high-accuracy numerical integration of Eq. (1). If $x_0 \sim p_{u_{\min}}$, the teacher marginal at coordinate u is the pushforward

$$p_u = (F_{u_{\min} \rightarrow u})_\# p_{u_{\min}}. \quad (2)$$

A fixed student solver $\mathcal{S}$ performs a one-step update $\Phi_{\mathcal{S}}(x; u \rightarrow v)$ that approximates $F_{u \rightarrow v}(x)$. Given a target number of steps N , the goal is to construct a schedule

$$U = (u_0 < u_1 < \dots < u_N), \quad u_0 = u_{\min}, \quad u_N = u_{\max}, \quad (3)$$

without retraining f_θ or changing the solver.



All stages are schedule calibration only; model weights stay fixed.

Figure 1: Our calibration pipeline. Teacher trajectories define oracle-start residual probes, distributional risk forms a monitor clock, and equal-monitor-mass inversion materializes a solver-aware schedule. The schematic contains method structure only.

134 3.2 Teacher trajectories sample teacher marginals

135 We draw K calibration initial states $x_{k,0} \sim p_{u_{\min}}$ and compute high-accuracy teacher trajectories

$$x_k^*(u) = F_{u_{\min} \rightarrow u}(x_{k,0}), \quad k = 1, \dots, K. \quad (4)$$

136 By Eq. (2), for every coordinate u the set $\{x_k^*(u)\}_{k=1}^K$ is a Monte Carlo sample from p_u . Calibration
137 therefore estimates a distributional object: the local state distribution on which the student solver is
138 evaluated.

139 3.3 Geometric oracle-start defect

140 We evaluate endpoint residuals with a positive definite state-space metric $G(u) \succ 0$:

$$\langle r_1, r_2 \rangle_{G(u)} = r_1^\top G(u) r_2, \quad \|r\|_{G(u)}^2 = r^\top G(u) r. \quad (5)$$

141 The metric can normalize errors across noise levels, channels, or latent-space scalings; examples
142 include isotropic noise normalization $G(u) = (\sigma(u)^2 + \sigma_{\text{data}}^2)^{-1} I$ or channel-wise whitening.

143 For trajectory k , define the G -unit teacher tangent

$$T_{k,G}(u) = \frac{\partial_u x_k^*(u)}{\|\partial_u x_k^*(u)\|_{G(u)}} = \frac{f_\theta(x_k^*(u), u)}{\|f_\theta(x_k^*(u), u)\|_{G(u)}}. \quad (6)$$

144 When the tangent norm is numerically unstable, we use the full residual by setting $\rho = 1$. For a
145 residual r measured at endpoint b , define

$$\|r\|_{\rho,k,b}^2 = r^\top G(b) r - (1 - \rho) (T_{k,G}(b)^\top G(b) r)^2, \quad \rho \in (0, 1]. \quad (7)$$

146 Equivalently, if $r = r_{\parallel}^{G(b)} + r_{\perp}^{G(b)}$ is the $G(b)$ -orthogonal tangent-normal decomposition, then

$$\|r\|_{\rho,k,b}^2 = \|r_{\perp}^{G(b)}\|_{G(b)}^2 + \rho \|r_{\parallel}^{G(b)}\|_{G(b)}^2. \quad (8)$$

147 Thus $\rho = 1$ gives the full residual, while $0 < \rho < 1$ downweights phase-like tangent error and
148 emphasizes normal deviation from the teacher trajectory.

149 For $x \sim p_u$, the oracle-start local prediction defect of the student solver over $[u, v]$ is

$$d_S(x; u, v) = \|\Phi_S(x; u \rightarrow v) - F_{u \rightarrow v}(x)\|_{\rho,u,v}^2. \quad (9)$$

150 On calibration trajectory k , this becomes

$$d_k(u, v) = \|\Phi_S(x_k^*(u); u \rightarrow v) - x_k^*(v)\|_{\rho,k,v}^2. \quad (10)$$

151 Every measurement starts from $x_k^*(u)$, not from a student state reached after previous approximate
152 steps. The defect therefore measures intrinsic one-step solver error rather than a mixture of local
153 defect and accumulated prefix error.

154 3.4 Local coefficient estimation and risk aggregation

155 For small $h > 0$, assume the local expansion

$$d_{\mathcal{S}}(x; u, u + h) = a_{\mathcal{S}}(x, u)h^q + O(h^{q+1}), \quad (11)$$

156 where $q > 0$ is the effective squared-defect order and $a_{\mathcal{S}}(x, u) \geq 0$ is a local defect coefficient.
 157 Along trajectory k , $d_k(u, u + h) = a_k(u)h^q + O(h^{q+1})$ with $a_k(u) = a_{\mathcal{S}}(x_k^*(u), u)$. Given a probe
 158 step η , a single-probe estimate is

$$\hat{a}_k(u) = \frac{d_k(u, u + \eta)}{\eta^q}. \quad (12)$$

159 When q is unknown, we use multiple probe sizes η_j and fit

$$\log d_k(u, u + \eta_j) \approx \log a_k(u) + q \log \eta_j \quad (13)$$

160 by pooled log-log regression over trajectories and probe grid points.

161 Let $\mathcal{R}$ be a risk functional over nonnegative random variables. The distributional defect coefficient is

$$\bar{a}_{\mathcal{R}}(u) = \mathcal{R}_{x \sim p_u}[a_{\mathcal{S}}(x, u)], \quad (14)$$

162 estimated empirically by

$$\hat{\bar{a}}_{\mathcal{R}}(u) = \mathcal{R}_{k=1}^K[\hat{a}_k(u)]. \quad (15)$$

163 We use mean risk, CVaR risk, or a mean-CVaR mixture:

$$\bar{a}_{\text{mean}}(u) = \frac{1}{K} \sum_{k=1}^K \hat{a}_k(u), \quad (16)$$

$$\bar{a}_{\text{CVaR}}(u) = \text{CVaR}_{\beta, K}[\hat{a}_k(u)], \quad (17)$$

$$\bar{a}_{\alpha, \beta}(u) = (1 - \alpha)\bar{a}_{\text{mean}}(u) + \alpha\bar{a}_{\text{CVaR}}(u). \quad (18)$$

164 Mean risk targets average teacher-student alignment; CVaR emphasizes high-error trajectories; the
 165 mixture is the default robust choice.

166 3.5 Monitor clock and schedule construction

167 For a small interval $[u, u + h]$, define the distributional local defect

$$\bar{D}_{\mathcal{R}}(u, u + h) = \mathcal{R}_{x \sim p_u}[d_{\mathcal{S}}(x; u, u + h)]. \quad (19)$$

168 The local model implies $\bar{D}_{\mathcal{R}}(u, u + h) = \bar{a}_{\mathcal{R}}(u)h^q + O(h^{q+1})$. Equalizing the leading local risk over
 169 intervals with lengths h_i requires $\bar{a}_{\mathcal{R}}(u_i)h_i^q = C$, so the sampling density should be proportional to
 170 $\bar{a}_{\mathcal{R}}(u)^{1/q}$. We define the monitor

$$\omega_{\mathcal{R}}(u) = (\bar{a}_{\mathcal{R}}(u) + \epsilon_a)^{1/q}, \quad (20)$$

171 where $\epsilon_a > 0$ is a numerical floor. Its interval mass is

$$\mathcal{M}_{\mathcal{R}}([a, b]) = \int_a^b \omega_{\mathcal{R}}(u) du. \quad (21)$$

172 Let $\Omega_{\mathcal{R}} = \int_{u_{\min}}^{u_{\max}} \omega_{\mathcal{R}}(u) du$ and define the normalized monitor clock

$$\tau_{\mathcal{R}}(u) = \frac{\int_{u_{\min}}^u \omega_{\mathcal{R}}(\xi) d\xi}{\int_{u_{\min}}^{u_{\max}} \omega_{\mathcal{R}}(\xi) d\xi}. \quad (22)$$

173 The final N -step schedule is the inverse-CDF discretization

$$u_m = \tau_{\mathcal{R}}^{-1}\left(\frac{m}{N}\right), \quad m = 0, \dots, N, \quad (23)$$

174 which gives every interval equal monitor mass.

175 **Calibration procedure.** Given calibration samples, the teacher solver, target solver, metric, tangent
176 weight, risk functional, probe grid, probe sizes, and floor, the procedure:

- 177 1. computes dense teacher trajectories $x_k^*(u)$;
- 178 2. evaluates oracle-start defects $d_k(u, u + \eta_j)$ on the probe grid;
- 179 3. estimates q and $a_k(u)$ from single probes or pooled log-log fits;
- 180 4. aggregates $\hat{a}_{\mathcal{R}}(u)$ across trajectories;
- 181 5. builds $\hat{\omega}_{\mathcal{R}}(u)$, optionally smooths $\log \hat{\omega}_{\mathcal{R}}$, and exports $u_m = \hat{\tau}_{\mathcal{R}}^{-1}(m/N)$.

182 The dominant calibration cost is $O(KMJ C_S)$, where M is the probe-grid size, J is the number
183 of probe sizes, and C_S is the cost of one student probe. Changing mean aggregation to CVaR or a
184 mixture does not require additional model evaluations. The implementation exports explicit timestep
185 or sigma bundles with solver, NFE, seed, pilot, cache, and calibration-cost metadata.

186 4 Theory

187 Our theory concerns deterministic ODE solvers with oracle-start local defects. The statement excludes
188 stochastic, ancestral, and history-dependent solver paths unless they are rewritten as deterministic
189 augmented-state systems. This section derives the monitor in Eq. (20), proves coordinate covariance,
190 and states the local minimax guarantee.

191 The result has the same structural shape as path-length equalization in Score-Optimal Diffusion
192 Schedules [Williams et al., 2024]: a local cost density induces a monitor, and the inverse cumulative
193 monitor yields a constant-speed traversal in that geometry. The difference is the density. In Score-
194 Optimal scheduling, the density is derived from score-corrector work along the diffusion path. Here
195 it is the q -th root of a distributional oracle-start solver-defect coefficient, so the monitor is tied to the
196 target solver and the teacher marginal distribution used during calibration.

197 **Assumption 1** (Local power law). *For $x \sim p_u$, the oracle-start defect satisfies*

$$d_S(x; u, u + h) = a_S(x, u)h^q + R(x, u, h), \quad (24)$$

198 *where $q > 0$, $a_S(x, u) \geq 0$, and $|R(x, u, h)| \leq C_R(x, u)h^{q+1}$.*

199 **Assumption 2** (Risk regularity). *The risk functional $\mathcal{R}$ is monotone, positive homogeneous, and*
200 *subadditive on nonnegative random variables, and $\mathcal{R}_{x \sim p_u}[C_R(x, u)] < \infty$.*

201 Expectation, CVaR, worst-case risk, and nonnegative mixtures of coherent risks satisfy this assump-
202 tion.

203 **Theorem 1** (Distributional local power law). *Under Assumptions 1 and 2,*

$$\bar{D}_{\mathcal{R}}(u, u + h) = \bar{a}_{\mathcal{R}}(u)h^q + O(h^{q+1}), \quad (25)$$

204 *where*

$$\bar{D}_{\mathcal{R}}(u, u + h) = \mathcal{R}_{x \sim p_u}[d_S(x; u, u + h)], \quad \bar{a}_{\mathcal{R}}(u) = \mathcal{R}_{x \sim p_u}[a_S(x, u)]. \quad (26)$$

205 *Proof.* For expectation, linearity gives $\bar{D}_{\mathbb{E}} = h^q \mathbb{E}[a_S] + \mathbb{E}[R]$, and the remainder is bounded by
206 $\mathbb{E}[C_R]h^{q+1}$. For coherent risks, positive homogeneity gives $\mathcal{R}[a_S h^q] = h^q \mathcal{R}[a_S]$, while monotonic-
207 ity and subadditivity bound the perturbation by $\mathcal{R}[|R|] \leq \mathcal{R}[C_R]h^{q+1}$. $\square$

208 **Theorem 2** (Correct monitor density). *Suppose*

$$\bar{D}_{\mathcal{R}}(u, u + h) = \bar{a}_{\mathcal{R}}(u)h^q + O(h^{q+1}) \quad (27)$$

209 *with $\bar{a}_{\mathcal{R}}(u) > 0$. The leading-order solution of*

$$\min_{\{h_i\}} \max_i \bar{a}_{\mathcal{R}}(u_i)h_i^q, \quad \sum_i h_i = L, \quad (28)$$

210 *is obtained by equalizing $\bar{a}_{\mathcal{R}}(u_i)h_i^q$ across intervals. The corresponding continuous monitor density*
211 *is*

$$\omega_{\mathcal{R}}(u) = \bar{a}_{\mathcal{R}}(u)^{1/q}. \quad (29)$$

212 *Proof.* At leading order, interval i has risk $E_i = \bar{a}_{\mathcal{R}}(u_i)h_i^q$. If two active intervals have $E_i > E_j$,
 213 decreasing h_i and increasing h_j by a sufficiently small amount preserves $\sum_i h_i = L$ and decreases
 214 the maximum risk. Thus all active intervals satisfy $\bar{a}_{\mathcal{R}}(u_i)h_i^q = C$, so $h_i = C^{1/q}\bar{a}_{\mathcal{R}}(u_i)^{-1/q}$. The
 215 local sampling density is therefore proportional to $\bar{a}_{\mathcal{R}}(u)^{1/q}$. $\square$

216 **Theorem 3** (Coordinate covariance of the monitor). *Let $v = g(u)$ be a smooth monotone reparameterization. Suppose the same physical local defect can be expressed as $\bar{D}_{\mathcal{R}}(u, u + h_u) \approx \bar{a}_u(u)h_u^q$
 217 in u -coordinates and $\bar{D}_{\mathcal{R}}(v, v + h_v) \approx \bar{a}_v(v)h_v^q$ in v -coordinates. Then*

$$\omega_v(v) dv = \omega_u(u) du, \quad (30)$$

219 where $\omega_u(u) = \bar{a}_u(u)^{1/q}$ and $\omega_v(v) = \bar{a}_v(v)^{1/q}$. Thus $\omega(u) du$ is a coordinate-covariant monitor
 220 one-form.

221 *Proof.* Small increments satisfy $h_v = |dv/du|h_u$ and $h_u = |du/dv|h_v$ to leading order. The
 222 physical local defect is invariant under reparameterization, so $\bar{a}_u(u)h_u^q = \bar{a}_v(v)h_v^q$. Substituting
 223 $h_u = |du/dv|h_v$ gives $\bar{a}_v(v) = \bar{a}_u(u)|du/dv|^q$. Taking q -th roots yields $\omega_v(v) = \omega_u(u)|du/dv|$,
 224 and multiplication by dv gives the claim. $\square$

225 **Lemma 1** (Exact equalization of monitor mass). *Let $\omega_{\mathcal{R}}(u) > 0$ be continuous on $[u_{\min}, u_{\max}]$,
 226 and let $\Omega_{\mathcal{R}} = \int_{u_{\min}}^{u_{\max}} \omega_{\mathcal{R}}(u) du$. For any N -interval schedule, define $M_i = \int_{u_i}^{u_{i+1}} \omega_{\mathcal{R}}(u) du$. Then
 227 $\max_i M_i \geq \Omega_{\mathcal{R}}/N$. Equality holds exactly when all interval masses are equal, which is achieved by
 228 $u_m = \tau_{\mathcal{R}}^{-1}(m/N)$.*

229 **Assumption 3** (Uniform monitor approximation). *There exists $C_E > 0$ such that for all sufficiently
 230 small intervals $[s, t]$,*

$$|\bar{D}_{\mathcal{R}}(s, t) - \mathcal{M}_{\mathcal{R}}([s, t])^q| \leq C_E |t - s|^{q+1}. \quad (31)$$

231 Assume also $0 < \omega_{\min} \leq \omega_{\mathcal{R}}(u) \leq \omega_{\max} < \infty$.

232 **Theorem 4** (Asymptotic minimax optimality). *Let U^{Ω} be the equal-monitor-mass schedule. Under
 233 Assumption 3,*

$$\max_i \bar{D}_{\mathcal{R}}(u_i^{\Omega}, u_{i+1}^{\Omega}) \leq \left(\frac{\Omega_{\mathcal{R}}}{N}\right)^q + C_1 N^{-(q+1)}. \quad (32)$$

234 Moreover, for any schedule U satisfying $h_{\max} \leq C_h/N$,

$$\max_i \bar{D}_{\mathcal{R}}(u_i, u_{i+1}) \geq \left(\frac{\Omega_{\mathcal{R}}}{N}\right)^q - C_2 N^{-(q+1)}. \quad (33)$$

235 Thus the equal-monitor schedule is asymptotically minimax optimal for local distributional risk up to
 236 $O(N^{-(q+1)})$.

237 *Proof sketch.* The equal-mass schedule has monitor mass $\Omega_{\mathcal{R}}/N$ on every interval. Since $\omega_{\mathcal{R}} \geq \omega_{\min}$,
 238 its interval lengths are $O(N^{-1})$, and the upper bound follows from the uniform approximation. For
 239 the lower bound, any schedule has at least one interval with monitor mass at least $\Omega_{\mathcal{R}}/N$. If
 240 $h_{\max} \leq C_h/N$, the approximation error on that interval is $O(N^{-(q+1)})$. Appendix A.3 gives the full
 241 proof. $\square$

242 The implementation uses the regularized monitor $\omega_{\epsilon}(u) = (\bar{a}_{\mathcal{R}}(u) + \epsilon_a)^{1/q}$ to avoid zero monitor
 243 regions. The same statements apply to the regularized coefficient, and the unregularized target is
 244 recovered as $\epsilon_a \rightarrow 0$ on regions where $\bar{a}_{\mathcal{R}}(u) > 0$. These results support teacher-oracle local risk
 245 only; perceptual quality and universal FID gains require separate empirical evidence.

246 5 Experiments

247 Our studies are organized by claim. Final CIFAR-10 claims require 50k generated samples for FID,
 248 NFE coverage that includes 10 and 20, and more than one seed for reliability language, with matched
 249 base, native-coordinate linear, Karras/EDM-style [Karras et al., 2022], project-owned offline-proxy,
 250 and our schedules. The retained CIFAR backend is PNDM [Liu et al., 2022] with the Euler solver.
 251 Text-to-image diagnostics require matched prompt, seed, model, solver, NFE, and guidance-scale

Table 1: Real PNDM/CIFAR-10 smoke gate for our implementation. The run uses Euler, NFE 4, seed 0, and 32 generated samples per schedule. FID is reported only to verify that scoring produced finite, non-empty outputs; it is not paper-grade evidence.

Schedule	NFE	Seed	Samples	FID
Base	4	0	32	295.77
ours	4	0	32	258.82

Table 2: PNDM/CIFAR-10 50k FID for the Euler solver. Rows report mean $\pm$ standard error over three seeds (0, 1, 2), base, linear native-coordinate, fixed Karras/EDM-style, authorized project-owned offline, and our schedules. FID is lower-is-better and each seed uses 50k generated images. The final two columns are paired ours-minus-baseline FID gaps; negative values favor ours.

NFE	base	linear	Karras	offline	ours	ours — offline	ours — Karras
10	20.57 $\pm$ 0.08	20.57 $\pm$ 0.08	25.84 $\pm$ 0.10	19.20 $\pm$ 0.09	17.54 $\pm$ 0.08	-1.659 $\pm$ 0.017	-8.302 $\pm$ 0.031
20	11.17 $\pm$ 0.07	11.20 $\pm$ 0.07	10.64 $\pm$ 0.09	10.88 $\pm$ 0.06	10.71 $\pm$ 0.07	-0.175 $\pm$ 0.059	0.072 $\pm$ 0.024

252 keys across base, published AYS [Sabour et al., 2024], and our schedules. The repository also
 253 contains STORK adapter tests and scheduler code [Tan et al., 2026], but this draft does not report
 254 paper-grade STORK quality results.

255 Table 1 is a pipeline gate, not a paper result. It verifies that the repository can materialize our schedule,
 256 generate samples for base and our schedule, compute non-empty FID rows, and write calibration-cost
 257 metadata. The 32-sample FID values cannot support any image-quality claim.

258 Table 2 records the current three-seed 50k CIFAR-10 run for NFE 10 and 20, including native-
 259 coordinate linear, fixed Karras/EDM-style, and authorized project-owned offline-proxy baselines.
 260 The offline-proxy rows come from the project-owned coordinate-search optimizer; they are not
 261 published AYS schedules. The result supports only a bounded controlled-benchmark statement for
 262 the PNDM/CIFAR-10 backend, Euler solver, these two NFE values, and the reported baseline set.
 263 Appendix Figure 2 shows the same aggregate with per-seed points. Ours improves FID over base,
 264 linear, and offline at both NFE values. The Karras-style grid is worse than ours at NFE 10 but slightly
 265 better at NFE 20, so the positive claim is scoped away from uniform superiority over fixed schedules.
 266 Appendix Figure 3 visualizes the NFE-20 schedule profiles for seed 0. Appendix Table 4 gives a
 267 narrow actual reuse check for the same backend: reusing the seed-0 schedule for seeds 1 and 2 has
 268 zero observed FID gap because the continuous schedules round to the same PNDM Euler integer
 269 timesteps. Appendix Tables 5 and 6 record the matched text-to-image CFG sweeps where published
 270 AYS schedules enter the retained experiments. SD1.5 is a negative case for our schedule under
 271 CLIPScore and ImageReward. SDXL is condition-dependent: at CFG 5.0 our schedule improves
 272 ImageReward over base but trails AYS, while at CFG 7.5 it improves the mean CLIPScore and
 273 ImageReward deltas against both base and AYS. These sweeps support CFG-matched coverage
 274 and boundary analysis; broad text-to-image quality improvement remains unsupported. Appendix
 275 Figure 6 shows the representative base/AYS/ours sigma-grid diagnostic for CFG 7.5 and seed 0.

276 Table 3 reports the current calibration-cost accounting. The CIFAR-10 gain over base comes with
 277 calibration cost that exceeds the evaluated generation batch by 4.24–8.49 times. The SD1.5 sweep
 278 has a much larger calibration-to-generation ratio for the small matched prompt batch and negative
 279 ImageReward deltas. The SDXL rows use a deliberately lighter calibration configuration and
 280 have lower cost ratios, but they still require schedule reuse across many matched prompt batches
 281 for amortization. These numbers support cost-aware reporting and leave deployment efficiency
 282 unsupported.

283 **Takeaway.** The three-seed CIFAR setting improves over base, native-linear, and authorized project-
 284 owned offline-proxy schedules at NFE 10 and 20. The Karras comparison is mixed and stronger at
 285 NFE 20. The matched SD1.5 and SDXL sweeps establish text-to-image coverage, but their automated
 286 metrics are condition-dependent and constrain the claims.

Table 3: Calibration cost accounting for current paper-facing runs. The quality column reports base-minus-ours FID for CIFAR-10 and ours-minus-base ImageReward for text-to-image, so positive values favor ours. Model-evaluation equivalents count retained calibration cost and the evaluated generation batch, excluding unrelated pilot experiments.

Setting	condition	quality vs. base	cal. evals (M)	gen. evals (M)	cal./gen.
CIFAR-10	NFE 10	+3.026 $\pm$ 0.019 FID	12.73	1.500	8.49 $\times$
CIFAR-10	NFE 20	+0.464 $\pm$ 0.005 FID	12.73	3.000	4.24 $\times$
SD1.5	CFG 5	-0.133 IR	25.46	0.003	8,487 $\times$
SD1.5	CFG 7.5	-0.084 IR	25.46	0.003	8,487 $\times$
SD1.5	CFG 10	-0.087 IR	25.46	0.003	8,487 $\times$
SDXL	CFG 5	+0.079 IR	0.21	0.003	68.62 $\times$
SDXL	CFG 7.5	+0.122 IR	0.21	0.003	68.62 $\times$

6 Limitations

Our method equalizes local distributional solver defect rather than final perceptual quality. It should therefore be interpreted as teacher-alignment calibration, not as a direct optimizer for FID, CLIPScore, ImageReward, or human preference. The matched SD1.5 and SDXL CFG sweeps are retained because they show this distinction concretely.

The minimax result is asymptotic and relies on local power-law behavior. Very small step counts such as $N = 4$ or $N = 8$ may fall outside the regime where the theory is tight. The theory also assumes a deterministic ODE solver with oracle-start defects; multistep and scheduler-history adapters require an augmented-state analysis before the theorem applies exactly.

Calibration adds compute before generation. A schedule is practically useful only when reuse amortizes the teacher and probe cost, so the experiments report model-evaluation-equivalent calibration cost separately from generation cost. Finally, the empirical monitor depends on the representativeness of the calibration trajectories, the teacher solver accuracy, the chosen metric $G(u)$, and the stability of the fitted order q .

7 Conclusion

Our method frames low-NFE schedule selection as inference-time distributional defect equalization. It offers a bounded way to adapt schedules to the model, solver, NFE, and condition without updating model weights. The theory supports asymptotic local minimax calibration for deterministic ODE samplers under a power-law oracle-start defect model. The main positive evidence is the controlled three-seed CIFAR-10 result against base, native-coordinate linear, and authorized project-owned offline-proxy schedules, with mixed evidence against a fixed Karras/EDM-style grid. The matched SD1.5 and SDXL CFG sweeps identify condition-dependent text-to-image behavior under automated metrics. A narrow CIFAR reuse check shows no FID penalty when the seed-0 schedule is reused for seeds 1 and 2 in the same rounded PNDM Euler execution grid. Broader positive reuse validation, release packaging, and asset-license completion remain open before submission readiness.

References

- Jack Hessel, Ari Holtzman, Maxwell Forbes, Ronan Le Bras, and Yejin Choi. CLIPScore: A reference-free evaluation metric for image captioning, 2022. URL <https://arxiv.org/abs/2104.08718>.
- Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models, 2020. URL <https://arxiv.org/abs/2006.11239>.
- Tero Karras, Miika Aittala, Timo Aila, and Samuli Laine. Elucidating the design space of diffusion-based generative models, 2022. URL <https://arxiv.org/abs/2206.00364>.

320 Alex Krizhevsky. Learning multiple layers of features from tiny images. Techni-
321 cal report, University of Toronto, 2009. URL [https://www.cs.toronto.edu/~kriz/](https://www.cs.toronto.edu/~kriz/learning-features-2009-TR.pdf)
322 [learning-features-2009-TR.pdf](https://www.cs.toronto.edu/~kriz/learning-features-2009-TR.pdf).

323 Luping Liu, Yi Ren, Zhijie Lin, and Zhou Zhao. Pseudo numerical methods for diffusion models
324 on manifolds. In *International Conference on Learning Representations*, 2022. URL [https://](https://openreview.net/forum?id=PlKWVd2yBkY)
325 openreview.net/forum?id=PlKWVd2yBkY.

326 Cheng Lu, Yuhao Zhou, Fan Bao, Jianfei Chen, Chongxuan Li, and Jun Zhu. DPM-Solver: A
327 fast ODE solver for diffusion probabilistic model sampling in around 10 steps, 2022. URL
328 <https://arxiv.org/abs/2206.00927>.

329 Cheng Lu, Yuhao Zhou, Fan Bao, Jianfei Chen, Chongxuan Li, and Jun Zhu. DPM-Solver++: Fast
330 solver for guided sampling of diffusion probabilistic models, 2025. URL [https://arxiv.org/](https://arxiv.org/abs/2211.01095)
331 [abs/2211.01095](https://arxiv.org/abs/2211.01095).

332 Dustin Podell, Zion English, Kyle Lacey, Andreas Blattmann, Tim Dockhorn, Jonas Muller, Joe
333 Penna, and Robin Rombach. SDXL: Improving latent diffusion models for high-resolution image
334 synthesis, 2023. URL <https://arxiv.org/abs/2307.01952>.

335 Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal,
336 Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever.
337 Learning transferable visual models from natural language supervision, 2021. URL [https://](https://arxiv.org/abs/2103.00020)
338 arxiv.org/abs/2103.00020.

339 Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Bjorn Ommer. High-
340 resolution image synthesis with latent diffusion models, 2022. URL [https://arxiv.org/abs/](https://arxiv.org/abs/2112.10752)
341 [2112.10752](https://arxiv.org/abs/2112.10752).

342 Amirmojtaba Sabour, Sanja Fidler, and Karsten Kreis. Align your steps: Optimizing sampling
343 schedules in diffusion models. In Ruslan Salakhutdinov, Zico Kolter, Katherine Heller, Adrian
344 Weller, Nuria Oliver, Jonathan Scarlett, and Felix Berkenkamp, editors, *Proceedings of the 41st*
345 *International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning*
346 *Research*, pages 42947–42975. PMLR, 21–27 Jul 2024. URL [https://proceedings.mlr.](https://proceedings.mlr.press/v235/sabour24a.html)
347 [press/v235/sabour24a.html](https://proceedings.mlr.press/v235/sabour24a.html).

348 Tim Salimans and Jonathan Ho. Progressive distillation for fast sampling of diffusion models, 2022.
349 URL <https://arxiv.org/abs/2202.00512>.

350 Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models, 2022. URL
351 <https://arxiv.org/abs/2010.02502>.

352 Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben
353 Poole. Score-based generative modeling through stochastic differential equations, 2021. URL
354 <https://arxiv.org/abs/2011.13456>.

355 Zheng Tan, Weizhen Wang, Andrea L. Bertozzi, and Ernest K. Ryu. STORK: Faster diffusion and
356 flow matching sampling by resolving both stiffness and structure-dependence. In *International*
357 *Conference on Learning Representations*, 2026. doi: 10.48550/arXiv.2505.24210. URL [https://](https://arxiv.org/abs/2505.24210)
358 arxiv.org/abs/2505.24210.

359 Daniel Watson, William Chan, Jonathan Ho, and Mohammad Norouzi. Learning fast samplers for
360 diffusion models by differentiating through sample quality, 2022. URL [https://arxiv.org/](https://arxiv.org/abs/2202.05830)
361 [abs/2202.05830](https://arxiv.org/abs/2202.05830).

362 Christopher Williams, Andrew Campbell, Arnaud Doucet, and Saifuddin Syed. Score-
363 optimal diffusion schedules. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Pa-
364 quet, J. Tomczak, and C. Zhang, editors, *Advances in Neural Information Process-*
365 *ing Systems*, volume 37, pages 107960–107983. Curran Associates, Inc., 2024. doi:
366 10.52202/079017-3428. URL [https://proceedings.neurips.cc/paper_files/paper/](https://proceedings.neurips.cc/paper_files/paper/2024/file/c34057f0cfb685eadf7d8d7cc99f734a-Paper-Conference.pdf)
367 [2024/file/c34057f0cfb685eadf7d8d7cc99f734a-Paper-Conference.pdf](https://proceedings.neurips.cc/paper_files/paper/2024/file/c34057f0cfb685eadf7d8d7cc99f734a-Paper-Conference.pdf).

368 Jiazheng Xu, Xiao Liu, Yuchen Wu, Yuxuan Tong, Qinkai Li, Ming Ding, Jie Tang, and Yuxiao Dong.
 369 ImageReward: Learning and evaluating human preferences for text-to-image generation, 2023.
 370 URL <https://arxiv.org/abs/2304.05977>.

371 Shuchen Xue, Zhaoqiang Liu, Fei Chen, Shifeng Zhang, Tianyang Hu, Enze Xie, and Zhenguo Li.
 372 Accelerating diffusion sampling with optimized time steps. In *Proceedings of the IEEE/CVF*
 373 *Conference on Computer Vision and Pattern Recognition*, 2024. doi: 10.48550/arXiv.2402.17376.
 374 URL <https://arxiv.org/abs/2402.17376>.

375 Wenliang Zhao, Lujia Bai, Yongming Rao, Jie Zhou, and Jiwen Lu. UniPC: A unified predictor-
 376 corrector framework for fast sampling of diffusion models, 2023. URL <https://arxiv.org/abs/2302.04867>.
 377

378 A Additional Theory Details and Proofs

379 This appendix supplies the proof details that are not central to the main paper narrative. All results
 380 concern oracle-start local defects for deterministic ODE solvers and support local risk calibration
 381 only.

382 A.1 Mixed residual decomposition

383 Let $T = T_{k,G}(b)$ satisfy $T^\top G(b)T = 1$. The $G(b)$ -orthogonal tangent projection of r is

$$r_{\parallel}^{G(b)} = (T^\top G(b)r)T, \quad (34)$$

384 and $r_{\perp}^{G(b)} = r - r_{\parallel}^{G(b)}$. Orthogonality gives

$$\|r\|_{G(b)}^2 = \|r_{\perp}^{G(b)}\|_{G(b)}^2 + \|r_{\parallel}^{G(b)}\|_{G(b)}^2, \quad (35)$$

385 with

$$\|r_{\parallel}^{G(b)}\|_{G(b)}^2 = (T^\top G(b)r)^2. \quad (36)$$

386 Substituting into Eq. (7) yields Eq. (8).

387 A.2 Equal monitor mass

388 For completeness we prove Lemma 1. The interval masses satisfy $\sum_{i=0}^{N-1} M_i = \Omega_{\mathcal{R}}$, so their
 389 maximum is at least their average. Equality holds exactly when all masses are equal. Since $\omega_{\mathcal{R}} > 0$,
 390 the cumulative monitor clock $\tau_{\mathcal{R}}$ is strictly increasing and invertible, and the inverse-CDF construction
 391 in Eq. (23) gives equal increments of normalized cumulative mass.

392 A.3 Full proof of asymptotic minimax optimality

393 For the equal-mass schedule U^Ω ,

$$\mathcal{M}_{\mathcal{R}}([u_i^\Omega, u_{i+1}^\Omega]) = \frac{\Omega_{\mathcal{R}}}{N}. \quad (37)$$

394 Since $\omega_{\mathcal{R}}(u) \geq \omega_{\min}$,

$$h_i^\Omega = u_{i+1}^\Omega - u_i^\Omega \leq \frac{\Omega_{\mathcal{R}}/N}{\omega_{\min}} = O(N^{-1}). \quad (38)$$

395 Using Assumption 3,

$$\bar{D}_{\mathcal{R}}(u_i^\Omega, u_{i+1}^\Omega) \leq \mathcal{M}_{\mathcal{R}}([u_i^\Omega, u_{i+1}^\Omega])^q + C_E (h_i^\Omega)^{q+1} \quad (39)$$

$$\leq \left(\frac{\Omega_{\mathcal{R}}}{N}\right)^q + C_1 N^{-(q+1)}. \quad (40)$$

396 Taking the maximum over intervals proves the upper bound.

Now consider any admissible schedule U with $h_{\max} \leq C_h/N$. Define $M_i = \mathcal{M}_{\mathcal{R}}([u_i, u_{i+1}])$. Since $\sum_i M_i = \Omega_{\mathcal{R}}$, there exists j such that $M_j \geq \Omega_{\mathcal{R}}/N$. Again by Assumption 3,

$$\bar{D}_{\mathcal{R}}(u_j, u_{j+1}) \geq M_j^q - C_E h_j^{q+1} \quad (41)$$

$$\geq \left(\frac{\Omega_{\mathcal{R}}}{N}\right)^q - C_E \left(\frac{C_h}{N}\right)^{q+1} \quad (42)$$

$$= \left(\frac{\Omega_{\mathcal{R}}}{N}\right)^q - C_2 N^{-(q+1)}. \quad (43)$$

Therefore the maximum interval defect of any admissible schedule is at least this lower bound.

A.4 Monte Carlo estimation of the mean monitor

Since $x_k^*(u) \sim p_u$, the random variables $a_k(u) = a_{\mathcal{S}}(x_k^*(u), u)$ are i.i.d. samples from the distribution of $a_{\mathcal{S}}(x, u)$ under p_u . If the variance is finite, the central limit theorem gives

$$\frac{1}{K} \sum_{k=1}^K a_k(u) - \mathbb{E}_{x \sim p_u}[a_{\mathcal{S}}(x, u)] = O_p(K^{-1/2}). \quad (44)$$

If coefficients are estimated from finite probes, write $\hat{a}_k(u) = a_k(u) + \delta_k(u)$ with $K^{-1} \sum_k \delta_k(u) = O(\varepsilon_{\text{probe}}(u))$. Then

$$\hat{a}_{\text{mean}}(u) - \bar{a}_{\text{mean}}(u) = O_p(K^{-1/2}) + O(\varepsilon_{\text{probe}}(u)). \quad (45)$$

A.5 Inverse-CDF stability

Let

$$F(u) = \int_{u_{\min}}^u \omega_{\mathcal{R}}(\xi) d\xi, \quad \hat{F}(u) = \int_{u_{\min}}^u \hat{\omega}_{\mathcal{R}}(\xi) d\xi. \quad (46)$$

If $\sup_u |\hat{F}(u) - F(u)| \leq \varepsilon_F$ and $\omega_{\mathcal{R}}(u) \geq \omega_{\min} > 0$, then F^{-1} is Lipschitz with constant at most $1/\omega_{\min}$. The corresponding exact and estimated schedules satisfy

$$\max_m |\hat{u}_m - u_m| \leq \frac{\varepsilon_F}{\omega_{\min}} + O(\varepsilon_F^2). \quad (47)$$

The second-order term accounts for local nonlinear curvature of the inverse map and normalization effects in the estimated cumulative mass.

A.6 Distributional coupling interpretation

Let $p_{*,T}$ be the terminal teacher distribution and $p_{S,T}$ the terminal student distribution induced by schedule U . For a shared initial sample x_0 , define

$$x_T^* = F_{u_{\min} \rightarrow T}(x_0), \quad x_T^S = \Phi_{S,U}(x_0). \quad (48)$$

This shared-noise construction is a coupling between $p_{*,T}$ and $p_{S,T}$, hence

$$W_2^2(p_{S,T}, p_{*,T}) \leq \mathbb{E}_{x_0} [\|x_T^S - x_T^*\|_{G_T}^2]. \quad (49)$$

Under a first-order error propagation expansion,

$$x_T^S - x_T^* = \sum_{i=0}^{N-1} S_i e_i + \text{higher-order terms}, \quad (50)$$

where e_i is the oracle-start local error at step i , stability of the downstream flow gives the conservative proxy

$$W_2^2(p_{S,T}, p_{*,T}) \leq C_{\text{stab}}(U) \sum_{i=0}^{N-1} \mathbb{E}_{x \sim p_{u_i}} [d_{\mathcal{S}}(x; u_i, u_{i+1})] + \text{higher-order terms}. \quad (51)$$

This is an interpretation, not the main guarantee: local defect controls a first-order coupling proxy under stability assumptions, but it is not an exact final image-quality objective.

Table 4: Actual seed-0 schedule reuse for PNDM/CIFAR-10 with `pndm_model_ddim_cifar10`, Euler solver, NFE 10 and 20, seeds 1 and 2, and 50k generated images per seed. FID is lower-is-better. The reuse gap is seed-0-reuse FID minus seed-specific FID for our schedule. For these rows the continuous schedules differ by less than 0.002 timestep units and round to the same integer PNDM Euler execution grids.

	NFE	base	per-seed ours	reused ours	reuse gap
	10	20.49 ± 0.02	17.47 ± 0.05	17.47 ± 0.05	0.000 ± 0.000
	20	11.10 ± 0.02	10.64 ± 0.02	10.64 ± 0.02	0.000 ± 0.000

B Reproducibility Notes

All paper-facing runs were executed on a local workstation with four NVIDIA GeForce RTX 4090 GPUs using a PyTorch/CUDA inference stack. The repository stores experiment configurations, cleaned result CSVs, plotting inputs, copied schedule metadata, and prompt manifests alongside the paper artifacts so that the tables and figures can be regenerated without relying on raw training logs.

The CIFAR-10 study uses the PNDM backend [Liu et al., 2022] with `pndm_model_ddim_cifar10`, Euler solver, NFE 10 and 20, and seeds 0, 1, and 2. Each reported FID row uses 50k generated images per seed. The baseline set includes the native base schedule, a native-coordinate linear schedule, a fixed Karras/EDM-style sigma grid [Karras et al., 2022], an authorized project-owned offline-proxy schedule, and our schedule. The offline baseline is an in-repository coordinate-search run for this CIFAR setting; it is not a published AYS CIFAR schedule. Additional lightweight and medium-budget offline-proxy attempts are retained only as failure evidence.

The text-to-image studies use latent-diffusion models [Rombach et al., 2022] with the Diffusers Euler solver, NFE 10, three random seeds, and 50 prompts from `diffusers_ablation_prompts`. Stable Diffusion 1.5 is evaluated at guidance scales 5.0, 7.5, and 10.0; SDXL is evaluated at guidance scales 5.0 and 7.5. Each comparison is matched by prompt, seed, model, solver, NFE, and guidance scale across base, published AYS [Sabour et al., 2024], and our schedule. We report CLIPScore [Hessel et al., 2022, Radford et al., 2021] and ImageReward [Xu et al., 2023] with pairwise mean deltas and win rates.

Schedule reuse is evaluated separately from schedule construction. For CIFAR-10, the seed-0 schedule is reused for seeds 1 and 2 under the same PNDM Euler execution grid and then re-evaluated with 50k generated images per seed. For SD1.5, the schedule calibrated at CFG 7.5 and seed 0 is reused across CFG 5.0, 7.5, and 10.0 and all three seeds. These checks measure reuse within the same model and solver family; they do not establish transfer across models, prompt distributions, or solver classes.

Calibration cost is reported in model-evaluation equivalents. For CIFAR-10 the quality column uses base-minus-ours FID, so positive values mean lower FID for our schedule. For text-to-image rows the quality column uses ours-minus-base ImageReward, so negative values mean lower ImageReward for our schedule. The anonymous archive omits raw AYS and authorized-offline numeric schedule bundles, but keeps cleaned metrics, summary tables, rendered figures, and copied paper-facing schedule metadata under `paper/results/`, `paper/tables/`, and `paper/figures/`.

C Three-seed 50k CIFAR-10 FID

The paper-scale CIFAR-10 run evaluates the PNDM backend [Liu et al., 2022] with `pndm_model_ddim_cifar10`, Euler solver, seeds 0, 1, and 2, NFE 10 and 20, and base, native-coordinate linear, fixed Karras/EDM-style [Karras et al., 2022], authorized project-owned offline-proxy, and our schedules. Each seed uses 50k generated images for FID. The table reports mean and standard error over seeds.

Figure 3 shows the NFE-20 schedule profiles for the seed-0 calibration artifact used in the table. The base and native-linear start grids overlap in this backend. Karras shifts more steps toward the low-noise end, while ours redistributes start timesteps toward the regions selected by the defect monitor.

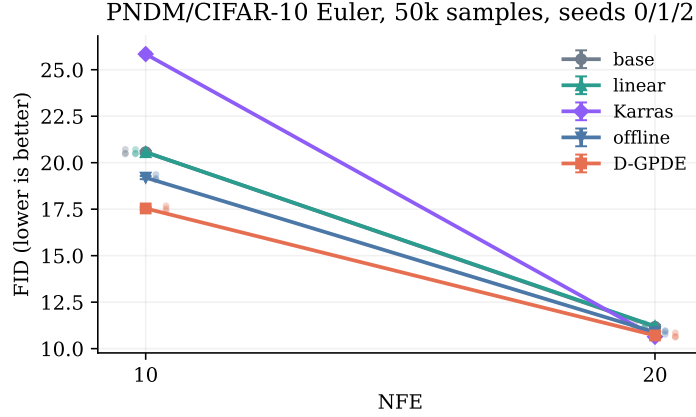


Figure 2: PNDM/CIFAR-10 50k FID for model `pndm_model_ddim_cifar10`, Euler solver, base, native-coordinate linear, fixed Karras/EDM-style, authorized project-owned offline-proxy, and our schedules, NFE 10 and 20, seeds 0, 1, and 2, final generation metric FID where lower is better. Error bars show standard error.

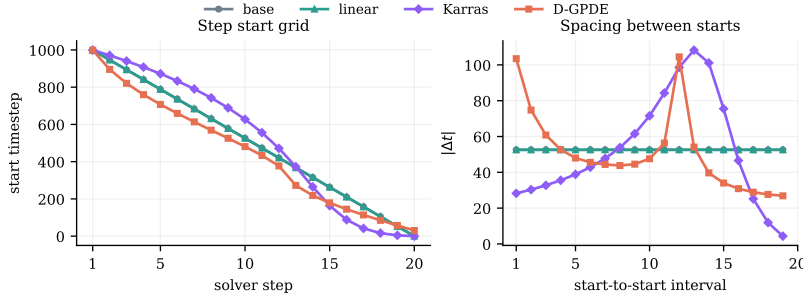


Figure 3: PNDM/CIFAR-10 Euler NFE-20 schedule profile for `pndm_model_ddim_cifar10`, seed 0. The left panel shows solver-step start timesteps for base, native-coordinate linear, Karras/EDM-style, and our schedules. The right panel shows spacing between consecutive start timesteps. This is a schedule diagnostic, not an additional image-quality metric.

461 Table 4 reports an actual generation reuse check: the seed-0 schedule bundle was loaded for seeds 1
 462 and 2, and FID was recomputed from 50k generated images per seed. In this PNDM Euler backend,
 463 the continuous seed-specific schedules differ from the seed-0 schedule by less than 0.002 timestep
 464 units. They round to the same integer execution timesteps, so the reuse check verifies no execution-
 465 level FID penalty for this configuration. It does not establish transfer across models, solvers, prompt
 466 distributions, or CFG conditions.

467 The cleaned metric rows, copied schedules, linear and Karras metadata, authorized of-
 468 fline metric rows, and oracle-reuse cost and reuse-validation tables are stored under
 469 `paper/results/cifar10_50k/`. The paper-facing archive omits authorized offline numeric sched-
 470 ule bundles. The source output roots are `outputs/gpde_pndm_cifar10_50k_nfe10_20_*`
 471 and `outputs/gpde_pndm_cifar10_50k_linear_baseline_seeds0_1_2/` and
 472 `outputs/gpde_pndm_cifar10_50k_karras_baseline_seeds0_1_2/` and
 473 `outputs/gpde_pndm_cifar10_authorized_offline/`; the actual reuse run lives under
 474 `outputs/gpde_pndm_cifar10_50k_reuse_seed0_schedule_seeds1_2/`.

475 D Matched text-to-image CFG sweeps

476 The text-to-image runs evaluate latent-diffusion models [Rombach et al., 2022] with the Diffusers
 477 Euler solver, NFE 10, seeds 0, 1, and 2, and 50 prompts from `diffusers_ablation_prompts`.

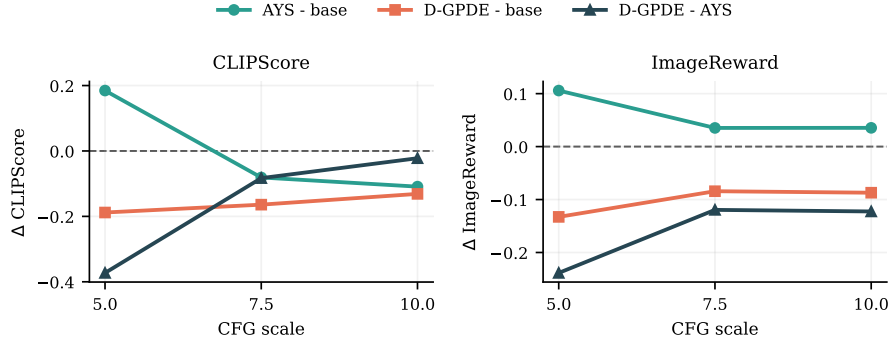


Figure 4: Matched SD1.5 text-to-image pairwise mean deltas for model `stable_diffusion_15`, Diffusers Euler solver, NFE 10, guidance scales 5.0, 7.5, and 10.0, schedules base, AYS, and ours, seeds 0–2, and 50 prompts per seed from `diffusers_ablation_prompts`. Deltas are computed over 150 matched prompt/seed pairs per guidance scale. CLIPScore and ImageReward are higher-is-better.

Table 5: SD1.5 text-to-image matched CFG sweep with Euler solver, NFE 10, seeds 0–2, and 50 prompts from `diffusers_ablation_prompts`. Rows report pairwise mean deltas and win rates over 150 matched prompt/seed pairs. CLIPScore and ImageReward are higher-is-better.

CFG	comparison	ΔCLIP	CLIP win (%)	ΔIR	IR win (%)
5	AYS - base	+0.18	53.3	+0.106	66.0
5	ours - base	-0.19	42.7	-0.133	33.3
5	ours - AYS	-0.37	41.3	-0.239	26.7
7.5	AYS - base	-0.08	51.3	+0.035	57.3
7.5	ours - base	-0.16	45.3	-0.084	46.0
7.5	ours - AYS	-0.08	47.3	-0.120	38.7
10	AYS - base	-0.11	46.0	+0.035	54.0
10	ours - base	-0.13	41.3	-0.087	40.7
10	ours - AYS	-0.02	48.0	-0.123	37.3

Each base, published AYS [Sabour et al., 2024], and our row is matched by prompt, seed, model, solver, NFE, and guidance scale. The metrics are CLIPScore [Hessel et al., 2022, Radford et al., 2021] and ImageReward [Xu et al., 2023], where higher is better.

D.1 Stable Diffusion 1.5

The SD1.5 sweep uses guidance scales 5.0, 7.5, and 10.0. It retains the largest text-to-image calibration setting in this draft, with 9 calibrated schedules and 1350 scored images.

D.2 SDXL

The SDXL sweep [Podell et al., 2023] uses guidance scales 5.0 and 7.5. It uses a lighter calibration configuration than SD1.5: one prompt per batch, four calibration batches, 64 reference steps, a 65-node reference grid, and a 128-node candidate grid. This choice keeps the run within the minimum validation budget while preserving the matched base/AYS/ours comparison across three seeds.

The SDXL result is condition-dependent. At CFG 5.0, our schedule improves ImageReward over base by 0.079 but trails AYS by 0.119. At CFG 7.5, it improves the mean CLIPScore and ImageReward deltas over both base and AYS. This pattern supports condition-aware evaluation coverage and keeps the text-to-image claim bounded to matched automated metrics.

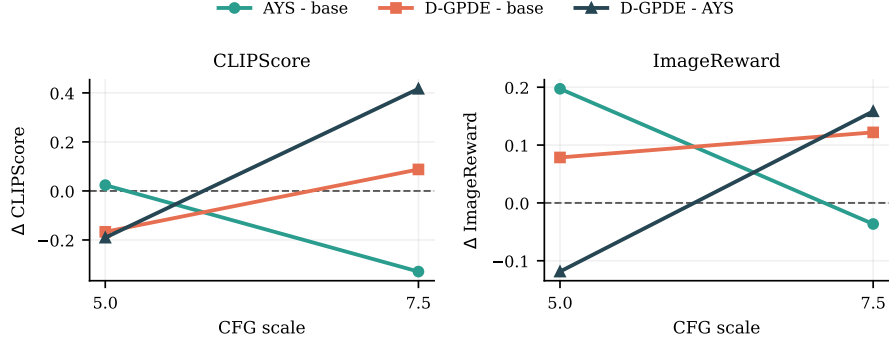


Figure 5: Matched SDXL text-to-image pairwise mean deltas for model sdx1, Diffusers Euler solver, NFE 10, guidance scales 5.0 and 7.5, schedules base, AYS, and ours, seeds 0–2, and 50 prompts per seed from `diffusers_ablation_prompts`. Deltas are computed over 150 matched prompt/seed pairs per guidance scale. CLIPScore and ImageReward are higher-is-better.

Table 6: SDXL text-to-image matched CFG sweep with Euler solver, NFE 10, seeds 0–2, and 50 prompts from `diffusers_ablation_prompts`. Rows report pairwise mean deltas and win rates over 150 matched prompt/seed pairs. CLIPScore and ImageReward are higher-is-better.

CFG	comparison	Δ CLIP	CLIP win (%)	Δ IR	IR win (%)
5	AYS - base	+0.02	46.0	+0.197	66.0
5	ours - base	-0.17	43.3	+0.079	52.7
5	ours - AYS	-0.19	48.0	-0.119	35.3
7.5	AYS - base	-0.33	44.7	-0.036	49.3
7.5	ours - base	+0.09	46.7	+0.122	54.7
7.5	ours - AYS	+0.42	58.0	+0.158	52.7

493 D.3 SD1.5 reuse and schedule diagnostics

494 Table 7 reports a second actual reuse check. We reuse the CFG 7.5, seed-0 schedule for CFG 5.0,
 495 7.5, and 10.0, seeds 0–2, with the same 50 prompts. Reuse stays near the CFG- and seed-specific
 496 rows on ImageReward, with matched deltas from +0.003 to +0.034, but loses CLIPScore by up
 497 to 0.10. Against base and AYS, reuse remains negative on both automated metrics in the pairwise
 498 summaries. The result broadens reuse evidence while preserving the text-to-image conclusion as
 499 mixed or negative.

500 Figure 6 shows the sigma-grid diagnostic for the CFG 7.5, seed-0 schedule bundle. Our schedule
 501 keeps larger sigma values for more early boundaries than both base and AYS in this representative
 502 case. This diagnostic explains allocation only; the quality metrics below remain the evidence for the
 503 mixed result.

504 The SD1.5 sweep is a retained mixed result. AYS improves ImageReward over base at all three
 505 guidance scales, while ours trails base and AYS on ImageReward and underperforms base on
 506 CLIPScore under this SD1.5/Euler/NFE-10 setup. This evidence supports a bounded failure anal-
 507 ysis and CFG-matched evaluation coverage. Cleaned rows, pairwise summaries, copied sched-
 508 ules, and oracle-reuse cost accounting are stored under `paper/results/t2i/`. The same direc-
 509 tory includes the prompt manifest and prompt-index CSV for `diffusers_ablation_prompts`.
 510 The source output roots are `outputs/gpde_diffusers_sd15_nfe10_cfg_seed_sweep/` and
 511 `outputs/gpde_diffusers_sdx1_nfe10_cfg_seed_sweep/`.

Table 7: Actual SD1.5 schedule reuse check. We reuse a single our schedule calibrated at CFG 7.5, seed 0 for CFG 5.0, 7.5, and 10.0, seeds 0–2, and 50 prompts per seed. Rows report matched prompt/seed deltas for reuse minus the CFG- and seed-specific schedule. Higher is better for both metrics.

CFG	Δ CLIP	CLIP win (%)	Δ ImageReward	IR win (%)
5	-0.10	46.7	+0.006	50.7
7.5	-0.05	28.0	+0.003	31.3
10	-0.00	48.7	+0.034	53.3

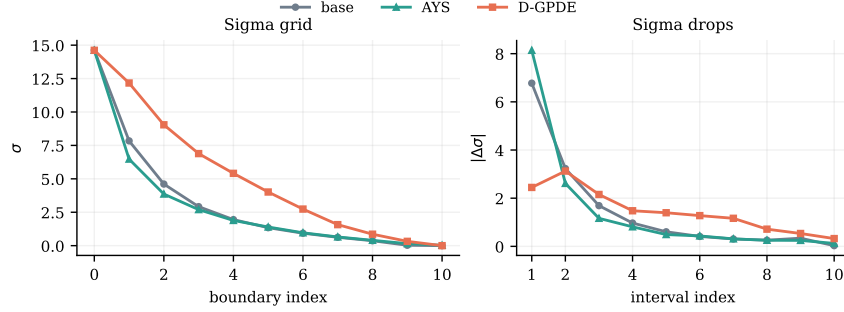


Figure 6: Stable Diffusion 1.5 Diffusers Euler NFE-10 schedule profile for guidance scale 7.5 and seed 0. The left panel shows the sigma grid for base, published AYS, and our schedules. The right panel shows sigma drops between adjacent boundaries. This is a schedule diagnostic, not a perceptual quality metric.

512 E Calibration Cost Accounting

513 Figure 7 visualizes the same cost rows as Table 3. The x-axis is logarithmic because the matched
 514 prompt batches are small relative to the per-guidance calibration cost. Positive quality values favor
 515 ours; negative values favor the base schedule under the reported metric.

516 The source CSV, table, and figure are generated by the aggregation script under
 517 `paper/figures/src/`. The cleaned summary is stored under `paper/results/cost/`.

518 Table 8 shows that the current CIFAR runs need schedule reuse across several evaluated-size batches
 519 before calibration overhead falls below generation cost. The actual CIFAR reuse check in Table 4
 520 supports this amortization only for the retained PNDM/CIFAR Euler execution grid. The SD1.5
 521 sweep would require thousands of matched prompt batches to amortize the calibration cost, and its
 522 observed ImageReward deltas are negative. The SDXL sweep uses a lighter calibration setting and
 523 breaks even after 69 matched prompt batches, but this arithmetic still needs positive reused-schedule
 524 quality before deployment efficiency becomes supported.

525 F Broader Impacts and Asset Notes

526 Our method is a schedule-calibration method, separate from model or dataset release. Its positive
 527 impact, when reuse is possible, is reduced generation error at a fixed NFE or fewer failed generations
 528 for a fixed model. Its main negative impact is that it can also make existing generative-image systems
 529 cheaper or easier to tune. Such systems inherit risks from their base models, including harmful
 530 content, social bias, mis- and disinformation, impersonation, and copyright or license misuse. The
 531 matched SD1.5 and SDXL results in this paper are reported as automated-metric boundary cases,
 532 leaving broad text-to-image quality and safer generation unsupported.

533 It also adds calibration compute. In settings where a calibrated schedule is used only once, to-
 534 tal compute can increase sharply. Any deployment claim should therefore account for sched-
 535 ule reuse, safety filtering, and model-license restrictions. This work uses no human subjects,

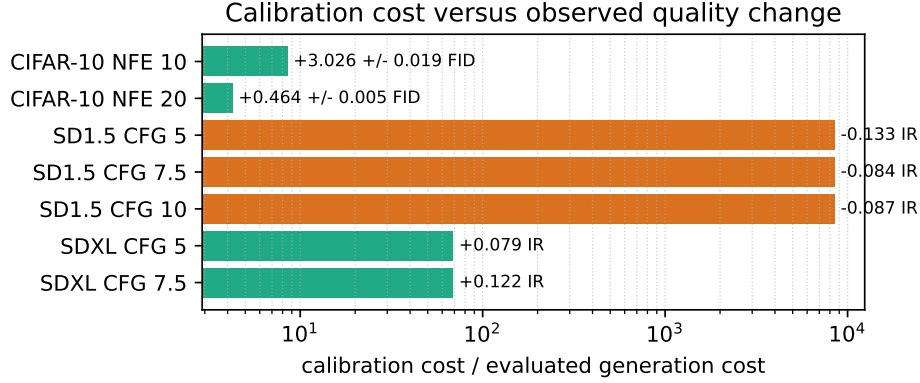


Figure 7: Calibration cost versus observed quality change for current paper-facing runs. CIFAR-10 uses base-minus-ours FID, so positive values mean lower FID for ours. Text-to-image rows use ours-minus-base ImageReward, so negative values mean lower ImageReward for ours. Model-evaluation equivalents count retained calibration cost divided by the evaluated generation batch cost.

Table 8: Calibration amortization under reuse of the same schedule. Columns report calibration overhead divided by generation cost after reusing the schedule for M evaluated-size batches. Break-even is the smallest integer M for which calibration cost is no larger than generation cost.

Setting	condition	quality vs. base	$M = 1$	$M = 10$	$M = 100$	break-even
CIFAR-10	NFE 10	+3.026 $\pm$ 0.019 FID	8.49 $\times$	0.85 $\times$	0.08 $\times$	9
CIFAR-10	NFE 20	+0.464 $\pm$ 0.005 FID	4.24 $\times$	0.42 $\times$	0.04 $\times$	5
SD1.5	CFG 5	-0.133 IR	8,487 $\times$	848.70 $\times$	84.87 $\times$	8487
SD1.5	CFG 7.5	-0.084 IR	8,487 $\times$	848.70 $\times$	84.87 $\times$	8487
SD1.5	CFG 10	-0.087 IR	8,487 $\times$	848.70 $\times$	84.87 $\times$	8487
SDXL	CFG 5	+0.079 IR	68.62 $\times$	6.86 $\times$	0.69 $\times$	69
SDXL	CFG 7.5	+0.122 IR	68.62 $\times$	6.86 $\times$	0.69 $\times$	69

private data, or deployed system. Complete local paths for the assets below are recorded in `configs/assets_manifest.yaml`.

The new assets produced by this paper are experiment configs, schedule metadata, cleaned CSVs, tables, plots, and manuscript text under `paper/`. Draft release metadata now lives under `paper/release/`, including an anonymized artifact manifest, license status note, and generated-sample policy. Before release, the authors still need to choose final licenses, finish the prompt redistribution audit, and prepare an external anonymous archive or repository. The current release policy omits raw AYS numeric schedule files from the anonymous package.

G Pilot CIFAR-10 result record

The project retains the existing PNDM/CIFAR-10 pilot rows because they are useful for debugging the claim-evidence loop. They are excluded from main-paper FID claims because the run uses one seed and 5k generated images, below the paper-grade 50k FID setting.

H Failure Analysis

The appendix preserves negative and mixed evidence. Retained records include schedule mismatch cases, solver adapter failures, teacher-alignment gains with generation-metric degradation, CFG mismatch, NaN or empty scoring checks, unstable timestep snapping, and settings where offline baselines remain stronger than ours.

Table 9: Current asset audit. “Pending” marks items that need final release documentation before submission.

Asset	Role	Source/license status	Local record
CIFAR-10	FID benchmark and reference statistics	Created by Krizhevsky, Nair, and Hinton; UCI lists CC BY 4.0; cite the technical report [Krizhevsky, 2009].	CIFAR config and FID-stat asset
PNDM CIFAR checkpoint	Controlled sampler backend	The official PNDM repository is Apache-2.0 and links the checkpoint/statistics folder; cite Liu et al. [2022]. The release manifest records the local CIFAR checkpoint and FID-stat hashes, but the archive should not bundle checkpoint files without final redistribution review.	PNDM CIFAR checkpoint asset
Stable Diffusion 1.5	Text-to-image backend	Cached model card lists <code>creativeml-openrail-m</code> .	SD1.5 cached model card
SDXL	Text-to-image backend	Cached model card and upstream terms need final author review.	SDXL cached model record
AYS 10-step schedule	Published offline schedule baseline	The official AYS quickstart lists the 10-step values; the release manifest records local hashes. The anonymous package omits raw numeric schedule files and retains AYS through citations, hashes, rendered figures, and metric rows.	AYS SD1.5 and SDXL 10-step bundles
Diffusers	Text-to-image execution library	Repository license is Apache-2.0.	Diffusers vendored checkout
ImageReward	Text-to-image metric	Model card and repository license are Apache-2.0.	ImageReward checkout
Prompt set	Matched text-to-image prompts	Project-local 50-prompt asset; the paper-facing manifest records hashes, while the release license still needs final audit.	SD prompt asset

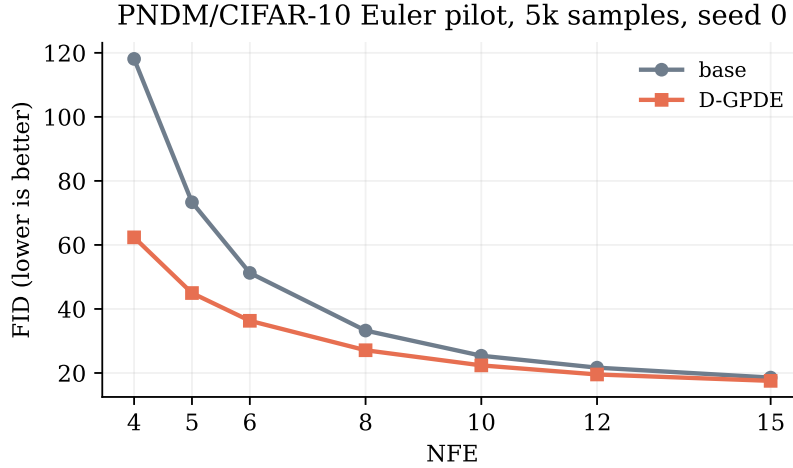


Figure 8: PNDM/CIFAR-10 pilot FID for model `pndm_model_ddim_cifar10`, Euler solver, base and our schedule schedules, NFE shown on the x-axis, seed 0, 5k generated images, final generation metric FID where lower is better. This is retained pilot evidence, not a paper-grade 50k FID result.

553 Current retained evidence includes PNDM rho/metric ablations, two failed project-owned offline-
554 proxy CIFAR diagnostics, and a matched SD1.5 CFG sweep. In the SD1.5 Euler NFE-10 sweep,
555 ours trails base and AYS on ImageReward across CFG 5.0, 7.5, and 10.0 and gives no CLIPScore
556 gain. This is a concrete negative text-to-image case.

557 Table 12 records a failed small-budget project-owned offline optimization attempt for PNDM/CIFAR-
558 10. The optimizer used the standalone hierarchical coordinate-search exporter with 256 training
559 images and no FID proxy early stopping. It is not the published AYS schedule. We retain it only as
560 negative evidence because its FID is much worse than base, Karras, and ours at both NFE values.

561 Before spending 50k samples on another failed schedule, we also ran a higher-budget offline-proxy
562 smoke diagnostic. The exporter used 1024 training images, seven candidates per coordinate, FID

Table 10: PNDM/CIFAR-10 FID for the Euler solver. Schedules are base and our schedule, NFE is listed per row, seed count is 1 seed (0), and FID is lower-is-better. Each row uses 5000 generated images. This is retained pilot evidence, not paper-grade 50k FID.

NFE	base FID ↓	ours FID ↓	FID reduction
4	118.11	62.37	55.74
5	73.33	44.97	28.36
6	51.26	36.30	14.95
8	33.25	27.10	6.14
10	25.39	22.38	3.01
12	21.67	19.54	2.13
15	18.56	17.54	1.02

Table 11: Retained failure and sensitivity cases. Positive FID deltas are worse; negative ImageReward deltas are worse.

Case	Setting	Comparison	Metric	Delta	Takeaway
rho=0 geometry	PNDM/CIFAR-10 Euler, NFE 10, seed 0, 5k samples	ours rho=0 - base	FID (lower better)	+30.922	Removing the residual mix makes FID much worse than base.
EDM scalar metric	PNDM/CIFAR-10 Euler, NFE 10, seed 0, 5k samples	ours EDM scalar - base	FID (lower better)	+4.871	This metric variant loses the base-schedule gain.
Mean aggregation	PNDM/CIFAR-10 Euler, NFE 15, seed 0, 5k samples	mean - CVaR	FID (lower better)	+0.071	Mean aggregation remains positive but trails CVaR.
SD1.5 CFG 5	SD1.5 Euler, NFE 10, 3 seeds, 50 prompts per seed	ours - base	ImageReward (higher better)	-0.133	Matched text-to-image calibration lowers ImageReward.
SD1.5 CFG 7.5	SD1.5 Euler, NFE 10, 3 seeds, 50 prompts per seed	ours - base	ImageReward (higher better)	-0.084	Matched text-to-image calibration lowers ImageReward.
SD1.5 CFG 10	SD1.5 Euler, NFE 10, 3 seeds, 50 prompts per seed	ours - base	ImageReward (higher better)	-0.087	Matched text-to-image calibration lowers ImageReward.

563 proxy early stopping for the 10-step stage, and 5k FID evaluation over seeds 0–2. It reached
564 26.76 ± 0.06 FID at NFE 10 and 18.32 ± 0.03 FID at NFE 20. These values remain far outside the
565 competitive range, so we keep the run as failure evidence. The paper stores the cleaned diagnostic
566 rows and optimizer config copy under `paper/results/failure/`.

567 We also logged a pre-stage-bundle default-budget attempt with 8192 training images and 11 candidates
568 per coordinate. Its 10-step proxy FID increased from 31.9996 to 75.8249 over four iterations after the
569 first best value, then the 20-step refinement stage remained slow before interruption. We therefore
570 treat that earlier run only as feasibility evidence for default-budget cost. The later authorized offline
571 run uses saved NFE 10 and 20 stage bundles and appears in the main CIFAR table. The paper stores
572 the cleaned one-row feasibility record under `paper/results/failure/`.

573 Figure 9 shows the three matched SD1.5 prompt/seed pairs with the largest ImageReward
574 drop for ours relative to the base Euler schedule. The paper stores the selection metadata in
575 `paper/results/failure/sd15_failure_grid_selection.csv`.

Table 12: Retained lightweight offline-proxy CIFAR-10 baseline. Rows report PNDM/CIFAR-10 Euler FID, mean $\pm$ standard error over seeds 0, 1, and 2 with 50k generated images per seed. The offline-proxy schedule is a small-budget project-owned hierarchical coordinate-search run, not the published AYS schedule. Lower FID is better.

NFE	base	Karras	ours	offline-proxy
10	20.57 ± 0.08	25.84 ± 0.10	17.54 ± 0.08	41.37 ± 0.13
20	11.17 ± 0.07	10.64 ± 0.09	10.71 ± 0.07	22.36 ± 0.05

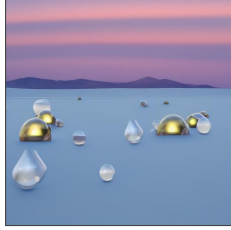
base
CFG 7.5, seed 1, prompt 46



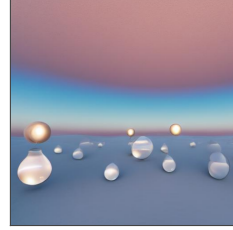
D-GPDE
Delta IR -2.45, Delta CLIP -1.46



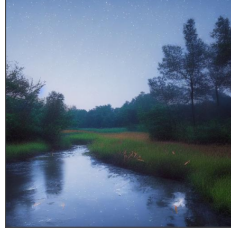
base
CFG 10, seed 0, prompt 18



D-GPDE
Delta IR -2.26, Delta CLIP -5.14



base
CFG 5, seed 1, prompt 38



D-GPDE
Delta IR -2.07, Delta CLIP -2.32

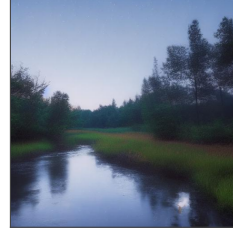


Figure 9: Qualitative failure grid for the matched SD1.5 Euler NFE-10 CFG sweep. Each row compares the base schedule with ours for the same prompt, seed, and guidance scale. The cases are selected by the largest ours-minus-base ImageReward drop among retained matched samples.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#).

Justification: The abstract and introduction explicitly bound the current claims to the method, the deterministic-ODE theory target, the three-seed CIFAR-10 result, and the mixed SD1.5 evidence.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes].

Justification: The paper includes a limitations section that separates deterministic-ODE theory scope, calibration cost, and teacher-alignment versus perceptual quality.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes].

Justification: Section 4 states the assumptions and theorem, while Appendix A provides the distributional local-risk reduction, coordinate-covariance argument, asymptotic local minimax proof, and inverse-CDF stability analysis.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.

- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes].

Justification: The reproducibility appendix lists the environment, hardware, exact launcher commands, aggregation commands, output locations, and manuscript build command for the current main experimental evidence.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No].

Justification: The repository contains code and scripts, and paper/release/ now builds a local draft archive, but the paper does not yet provide an external anonymized archive or author-approved release license.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes].

Justification: The result tables and appendix captions state model, solver, schedule, NFE, seeds, sample counts or prompt counts, guidance scales, prompt asset, and metric direction for the current evidence.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes].

Justification: The CIFAR-10 table and figure report standard error over three seeds for the main positive result, and the text avoids statistical-significance language beyond the retained uncertainty.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.

- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [No].

Justification: The appendix reports the local GPU type and memory and the paper reports model-evaluation-equivalent calibration cost, but complete wall-clock time and total project compute are not yet documented for every experiment.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines?>

Answer: [Yes].

Justification: The current work reports algorithmic experiments on public benchmark/model assets and does not involve human subjects, private data, or deployed systems.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes].

Justification: Appendix F discusses the possible efficiency benefit, inherited generative-model risks, misuse and bias concerns, calibration compute, and release constraints.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.

- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A].

Justification: The paper does not release a new pretrained image generator, language model, or scraped dataset; it evaluates schedules for existing public models.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [No].

Justification: Appendix F and paper/release/ record the current asset audit, including PNDM source URLs, local hashes, and a dated AYS/prompt source review. The authors decided to omit raw AYS numeric schedule files from the anonymous package; prompt text redistribution and paper-produced artifact licenses remain open.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.

- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [No].

Justification: Appendix F identifies the new configs, schedules, tables, figures, and result artifacts, and paper/release/ now provides draft release metadata plus a local draft archive. A final author-approved license and external anonymous package are still missing.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A].

Justification: The current experiments use automated metrics and do not involve crowdsourcing or human subjects.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A].

Justification: The current experiments do not involve human subjects, so IRB approval is not applicable.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.

- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. Declaration of LLM usage

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [N/A].

Justification: LLMs are not a core component of our method or its experimental evaluation; any writing or tooling assistance is outside the scientific method.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.